

Some problems in coarse graph theory

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Coarse graph theory is a new area concerned with the large-scale geometric structure of a graph.

Its origins lie in geometric group theory, where groups are frequently studied by looking at the large-scale structure of their Cayley graphs. But this perspective also gives ways of thinking about large-scale geometry of graphs without intrinsic algebraic structure.

A graph G naturally corresponds to a metric space $(V(G), d_G)$. Isomorphic graphs give isometric spaces.

A rougher idea of similarity is given by *quasi-isometry*. Quasi-isometry blurs out local structure, while keeping global geometry.

Definition

(X, d_X) is (L, C) -**quasi-isometric** to (Y, d_Y) if there is a map $f : X \rightarrow Y$ such that

- for all $x, y \in X$

$$\frac{1}{L}d_X(x, y) - C \leq d_Y(f(x), f(y)) \leq Ld_X(x, y) + C$$

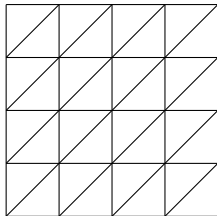
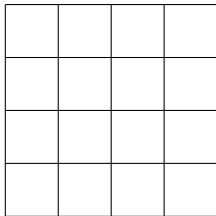
- for every $y \in Y$ there is $x \in X$ such that

$$d_Y(f(x), y) \leq C.$$

The first condition says that the metric is not distorted much at large scale. The second condition says that the mapping must be quasi-surjective (for example the natural inclusion $\mathbb{Z} \rightarrow \mathbb{Z}^2$ is not a quasi-isometry).

Examples:

- The grid and the grid with diagonals are quasi-isometric.

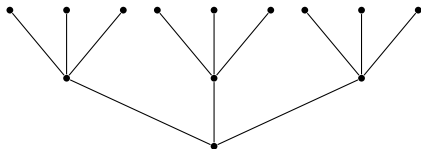
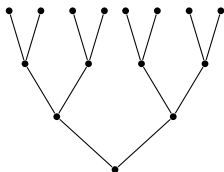


The identity is a quasi-isometry. So are

$$(x, y) \mapsto (2x, 3y)$$

$$(x, y) \mapsto (\lfloor x/2 \rfloor, \lfloor y/2 \rfloor)$$

- Any two d -dimensional normed spaces are quasi-isometric (via any linear bijection).
- The binary tree is quasi-isometric to the ternary tree.



- An (L, C) -quasi-isometry with $C = 0$ is a bi-Lipschitz map. If $L = 1$ we get additive distortion.

The importance of quasi-isometry in geometric group theory comes from influential work of Gromov, looking at the large-scale geometric structure of groups.

If G is a finitely generated group then any two Cayley graphs (with finite generating set) for G are quasi-isometric: for example, $(\mathbb{Z}, \{1\})$ and $(\mathbb{Z}, \{1, 2\})$.

So properties of Cayley graphs that are invariant under quasi-isometry (such as ends, asymptotic growth, ...) can be regarded as properties of the group, and we can study the large-scale geometry of a group by looking at its Cayley graphs.

Big question:

What does graph theory look like if we blur out local structure?

It was initially thought that much of classical graph theory would carry across, albeit with harder proofs. Indeed, an important paper of Geogakopoulos and Papasoglu conjectured coarse analogues for many classical graph-theoretic results.

But it turns out that life is more interesting:

- coarse graph theory has interesting new and different features
- graphs offer a richer playground than more symmetric structures such as Cayley graphs or (quasi-)transitive graphs

I will focus on two topics:

- coarse analogues of Menger's Theorem
- coarse analogues of graph minors

Let's start by introducing one question of each type.

Theorem (Menger, 1927)

Let $k \geq 1$ and $S, T \subseteq V(G)$. Then either there exist k vertex-disjoint S - T paths or there exists a set X of at most $k - 1$ vertices such that $G \setminus X$ has no S - T path.

But what if we want the paths to be far apart? Two groups (Georgakopoulos and Papasoglu; Albrechtsen, Huynh, Jacobs, Knappe and Wollan) conjectured:

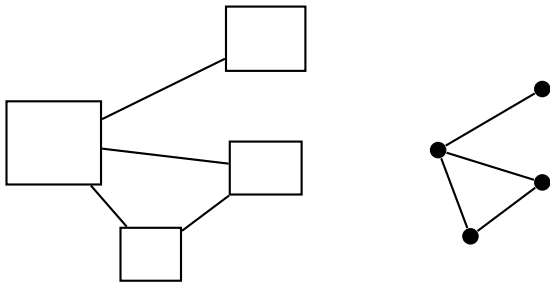
Conjecture (Coarse Menger Conjecture)

For all k, d there exists L such that for all G, S, T either there exist k S - T paths, pairwise distance $\geq d$ or there exists a set X of $k - 1$ balls of radius $\leq L$ such that $G \setminus X$ has no S - T path.

In other words, either there are k far apart S - T paths, or $k - 1$ balls that separate S from T .

Let's discuss graph minors.

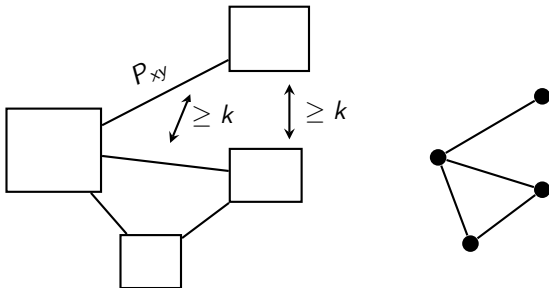
H is a minor of G if it can be obtained by deleting vertices and edges and contracting edges.



Graph minors are quite well understood: there are remarkably precise extremal results, and the Graph Minor Structure Theorem of Robertson and Seymour provides deep structural understanding.

What about a coarse analogue?

In this context *fat minors* were introduced by Bonamy, Bousquet, Esperet, Groenland, Liu, Pirot and Scott (an essentially equivalent notion was considered earlier by Chepoi, Dragan, Newman, Rabinovich and Vaxés).



In a k -fat minor, every two parts of the picture need to be at distance at least k , unless they are required to intersect.

What do graphs with no fat H -minor look like on the large scale? Is their structure similar to graphs with no H -minor?

Georgakopoulos and Papasoglu conjectured a positive answer:

Conjecture (Fat Minor Conjecture)

If G has no K -fat minor H then it is quasi-isometric to a graph with no H minor.

The Coarse Menger Conjecture and the Fat Minor Conjecture both assert that the coarse (large-scale) picture is similar to the classical picture. But is this the case?

The Coarse Menger Conjecture is known to be true in some cases. For example, both groups (Georgakopoulos and Papasoglu; Albrechtsen, Huynh, Jacobs, Knappe and Wollan) proved it for two paths.

And two groups (Gartland, Korhonen and Lokshtanov; and Hendrey, Norin, Steiner, and Turcotte) proved the following:

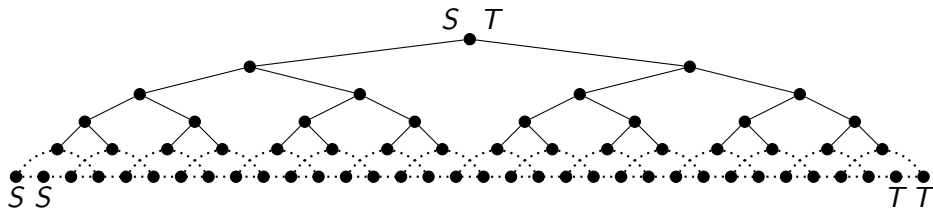
Theorem

*Let $k, D \geq 1$. There exists L such that for every graph with maximum degree at most D and all S, T , either there exist k S - T paths, pairwise distance **at least 2**, or there is a set of **at most $C(k, D)$ vertices** that separates S and T .*

What about the main conjecture?

Theorem (Nguyen, S., Seymour (2025))

The Coarse Menger Conjecture is false.



The dotted curves represent long paths.

A more elaborate construction (Nguyen, S., Seymour 2026+) shows that even weak versions of the conjecture, where $f(k, d)$ balls are allowed, are false.

So the Coarse Menger Conjecture is false in general. However, we have shown that it holds in some cases:

- bounded pathwidth
- treewidth 2
- when S and T lie on the boundary of a disc

The conjecture might hold for distance 2 (the induced case), and it might also hold on a fixed surface.

It would be very interesting to resolve it for planar graphs.

What about the Fat Minors Conjecture? Are H -fat-minor free graphs quasi-isometric to H -minor-free graphs?

This is known for a number of special cases: cycles, $K_{2,t}$, $K_4, \dots$

But the Fat Minors Conjecture is *false* in general: Dvořák, Hickingbotham, Illingworth and McCarty used the coarse Menger construction to build a counterexample.

This opens up the question: *what will a coarse graph minor theory look like?*

Some classical results survive in the coarse setting, while others need adjustment.

For example, there may be a *Coarse Kuratowski Theorem*: Georgakopoulos and Papasoglu conjectured that a graph with no fat K_5 or $K_{3,3}$ is quasi-isometric to a planar graph.

What about coarse analogues of *treewidth* and *pathwidth*? In the Graph Minor world, it is known that:

- Graphs of large treewidth have large grid minors
- Graphs of large pathwidth have large binary tree minors

What happens in the coarse setting? Davies and Albrechtsen recently used a variant of the coarse Menger counterexample to show that the naive coarse analogue of the Grid Minor Theorem does not hold.

But for pathwidth there is a positive result:

Theorem (Nguyen, S., Seymour)

Graphs with large coarse pathwidth contain fat minors of large binary trees.

Finding a suitable conditions for a coarse Grid Minor Theorem remains an interesting challenge!